

AIM

TO PERFORM PHASE IDENTIFICATION AND PHASE FRACTION DETERMINATION OF A GIVEN POLYCRYSTALLINE SAMPLE USING FULLPROF SOFTWARE, X-RAY DIFFRACTION (XRD) DATA AND RIETVELD REFINEMENT TECHNIQUE.

FullProf Software

FULLPROF IS A SOFTWARE USED FOR RIETVELD REFINEMENT OF X-RAY AND NEUTRON POWDER DIFFRACTION DATA.

IT HELPS REFINE CRYSTAL STRUCTURES BY FITTING THEORETICAL PATTERNS TO EXPERIMENTAL DATA.

USERS CAN ADJUST PARAMETERS LIKE LATTICE CONSTANTS, ATOMIC POSITIONS, AND THERMAL FACTORS.

IT'S WIDELY USED IN MATERIALS SCIENCE AND CRYSTALLOGRAPHY FOR STRUCTURAL ANALYSIS.

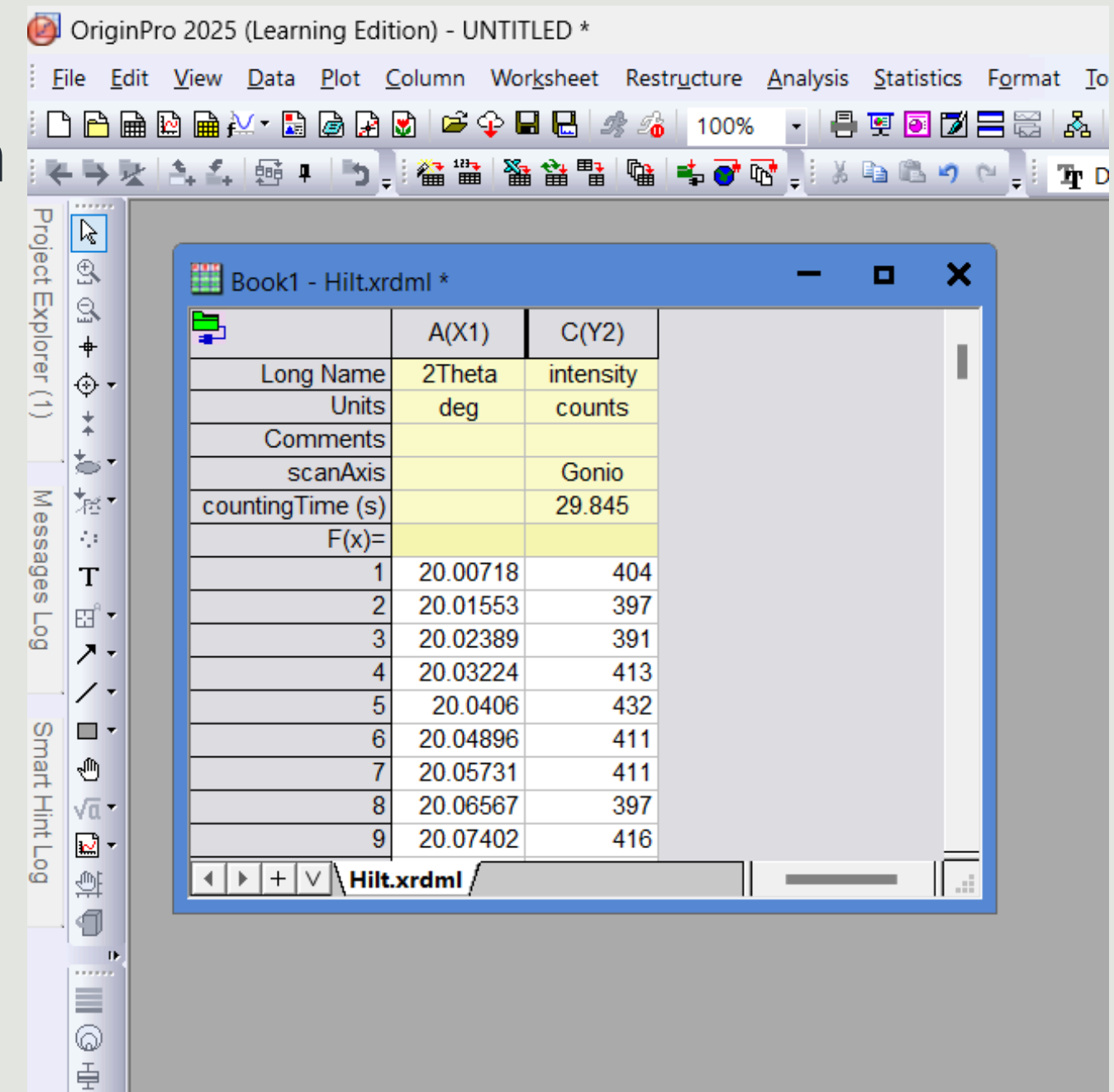
STEP-WISE RIETVELD REFINEMENT PROCESS

- **Obtain CIF File**

- Search COD database for your material's CIF file. Download it.

- **generate .Dat file**

- Open .xrdml data in Origin (or similar software).
- Export 2θ and Intensity (I) as a 2-column table (2θ , I).
- Save as data.dat (include 2θ -min, 2θ -max, step size in header/comments).



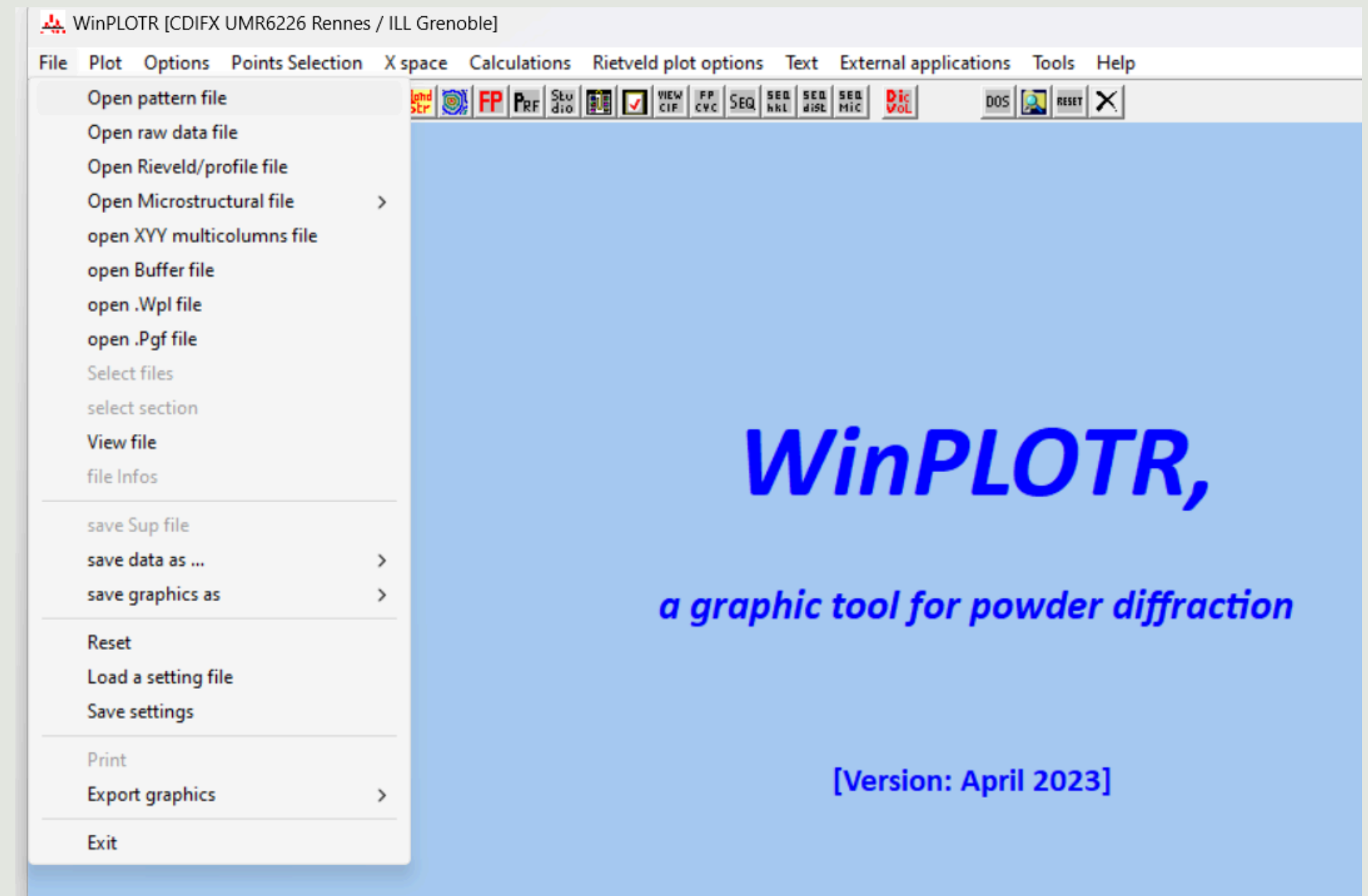
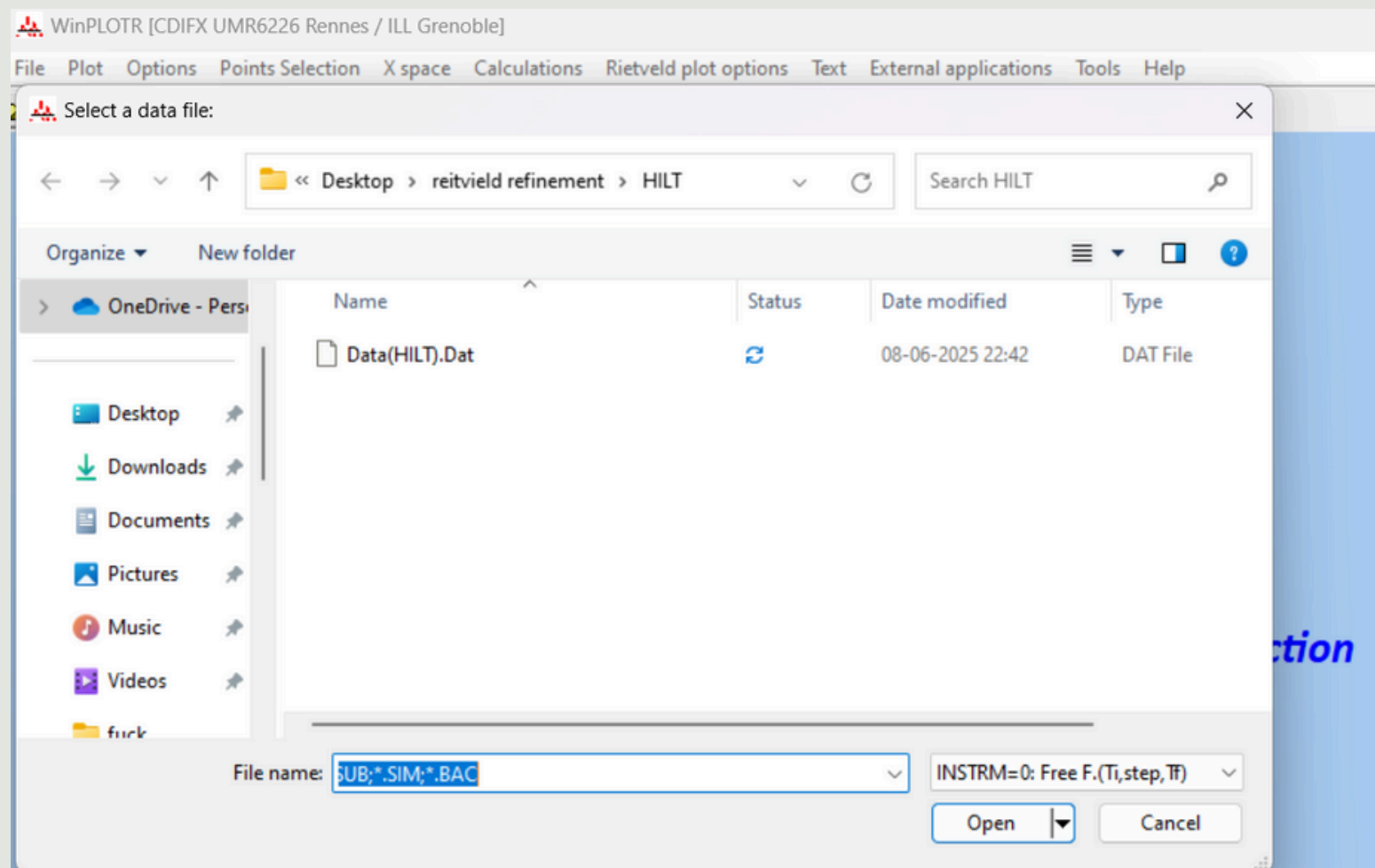
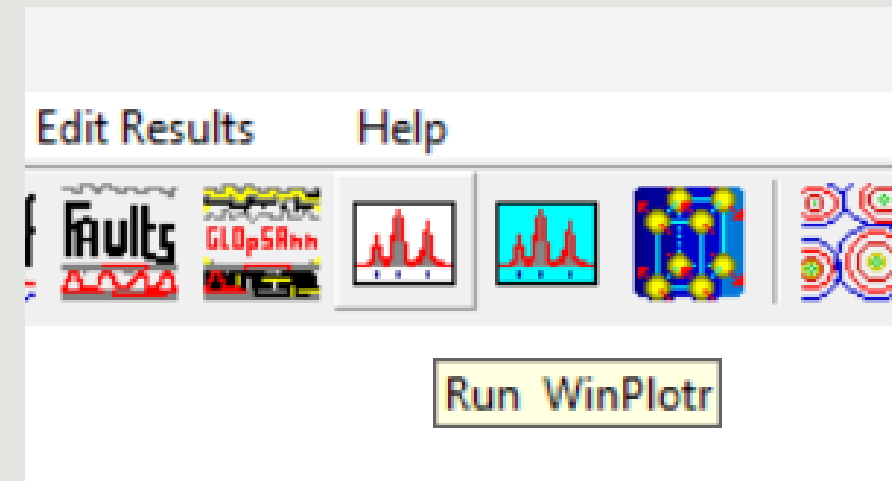
The screenshot shows the OriginPro 2025 (Learning Edition) interface. The main window displays a data table titled "Book1 - Hilt.xrdml *". The table has three columns: "A(X1)", "C(Y2)", and a third column for the data. The data is organized into a header section and a table of values.

	A(X1)	C(Y2)
Long Name	2Theta	intensity
Units	deg	counts
Comments		
scanAxis		Gonio
countingTime (s)		29.845
F(x)=		
1	20.00718	404
2	20.01553	397
3	20.02389	391
4	20.03224	413
5	20.0406	432
6	20.04896	411
7	20.05731	411
8	20.06567	397
9	20.07402	416

STEP-WISE RIETVELD REFINEMENT PROCESS

Generate Background File (.bgr)

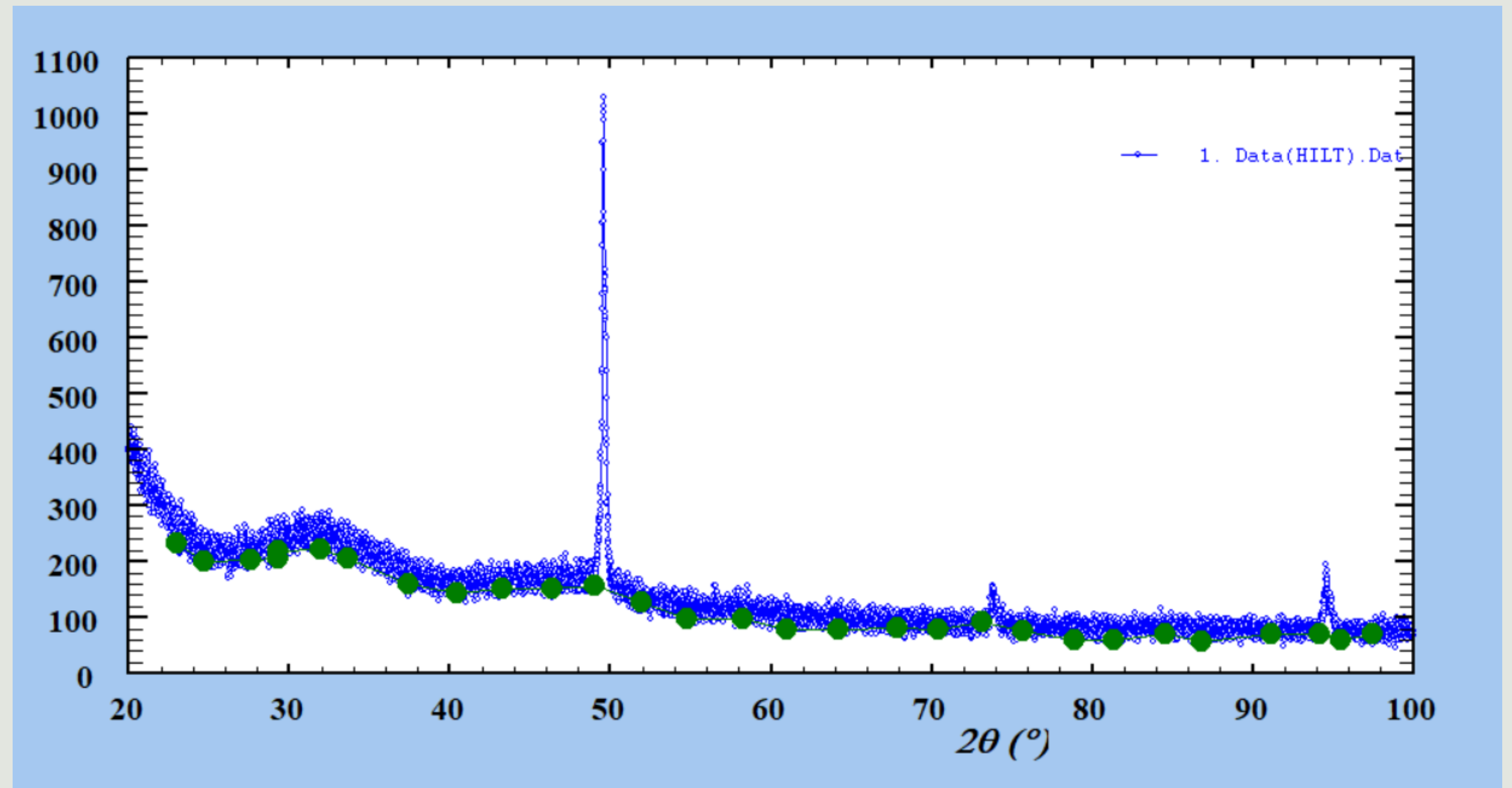
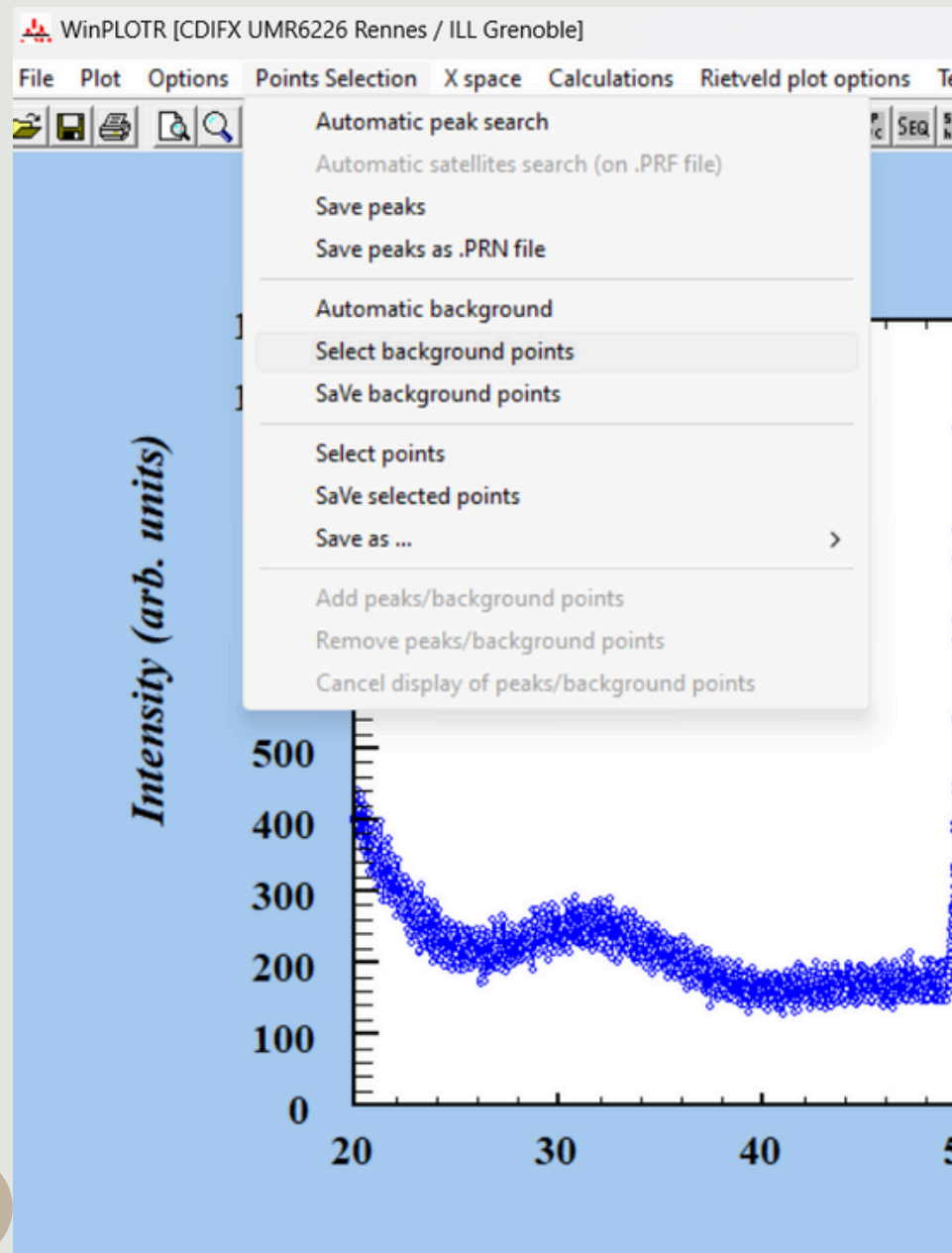
- Open data.dat in WinPlotr (FullProf suite).
- select open pattern file.
- select the format of data file as `INSTRM=0: Free F.(Ti,step,Tf)`
- select your .dat file.



STEP-WISE RIETVELD REFINEMENT PROCESS

Generate Background File (.bgr)

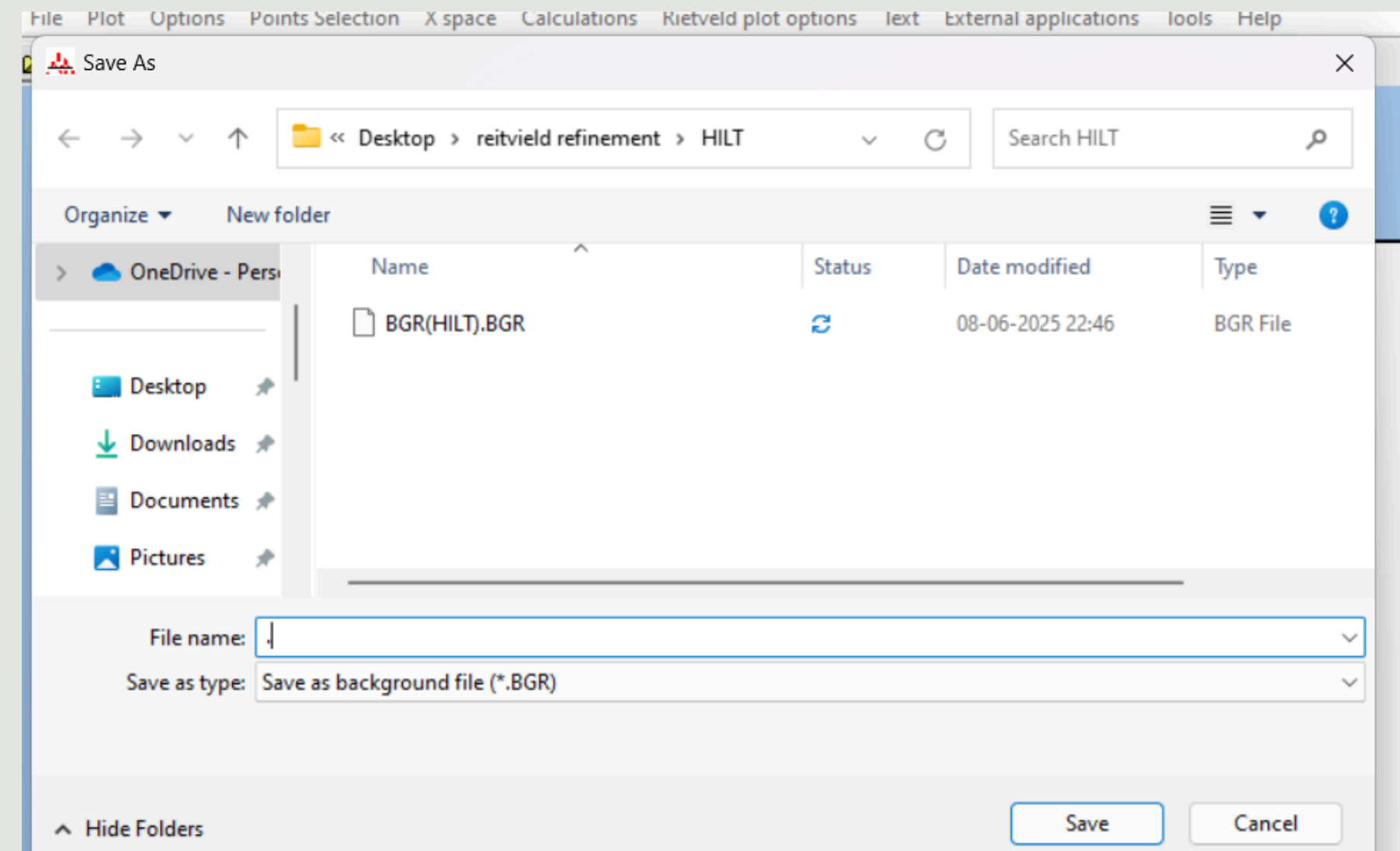
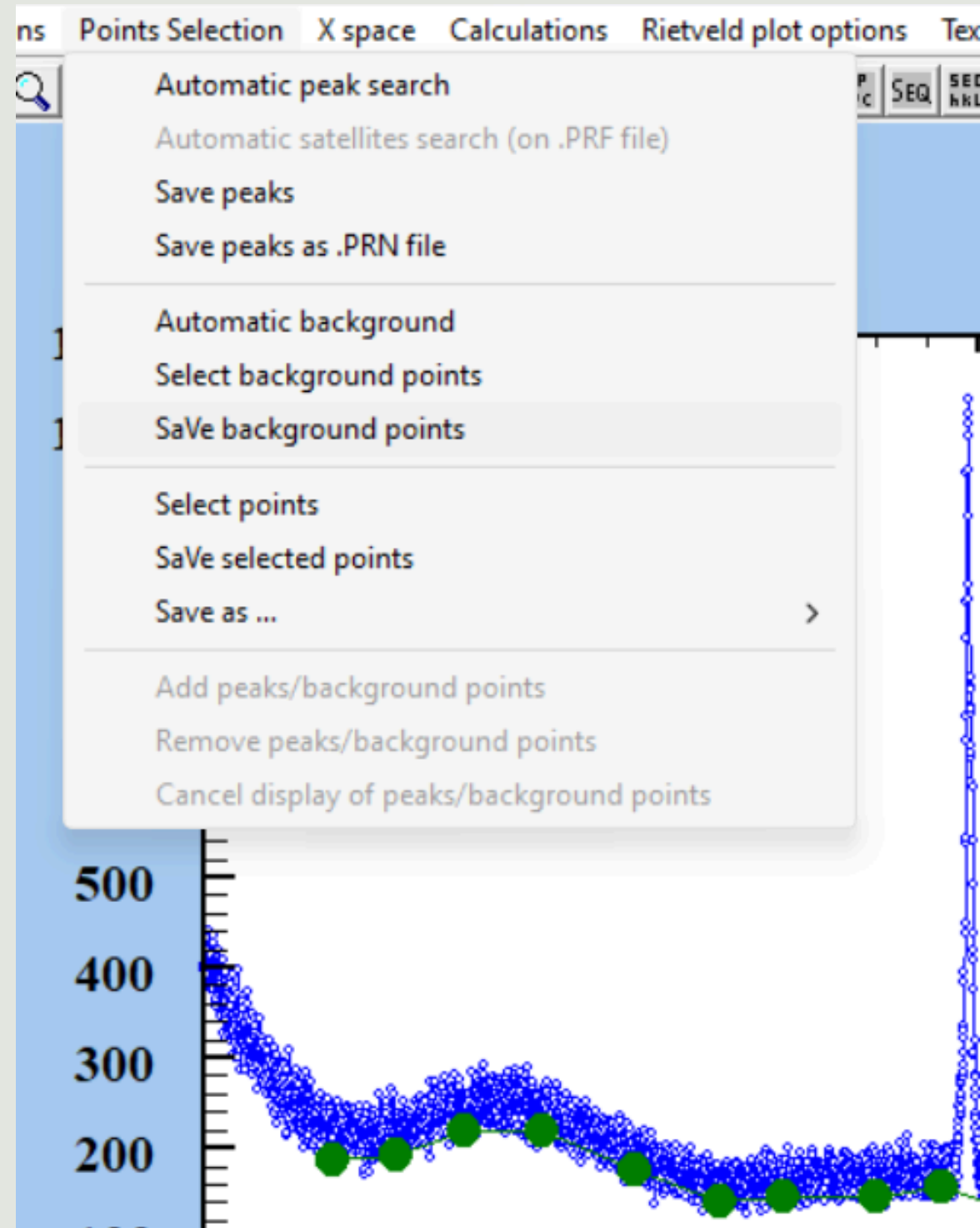
- go to points selections and Select background points manually on the pattern.



STEP-WISE RIETVELD REFINEMENT PROCESS

Generate Background File (.bgr)

- Save points as background.bgr.



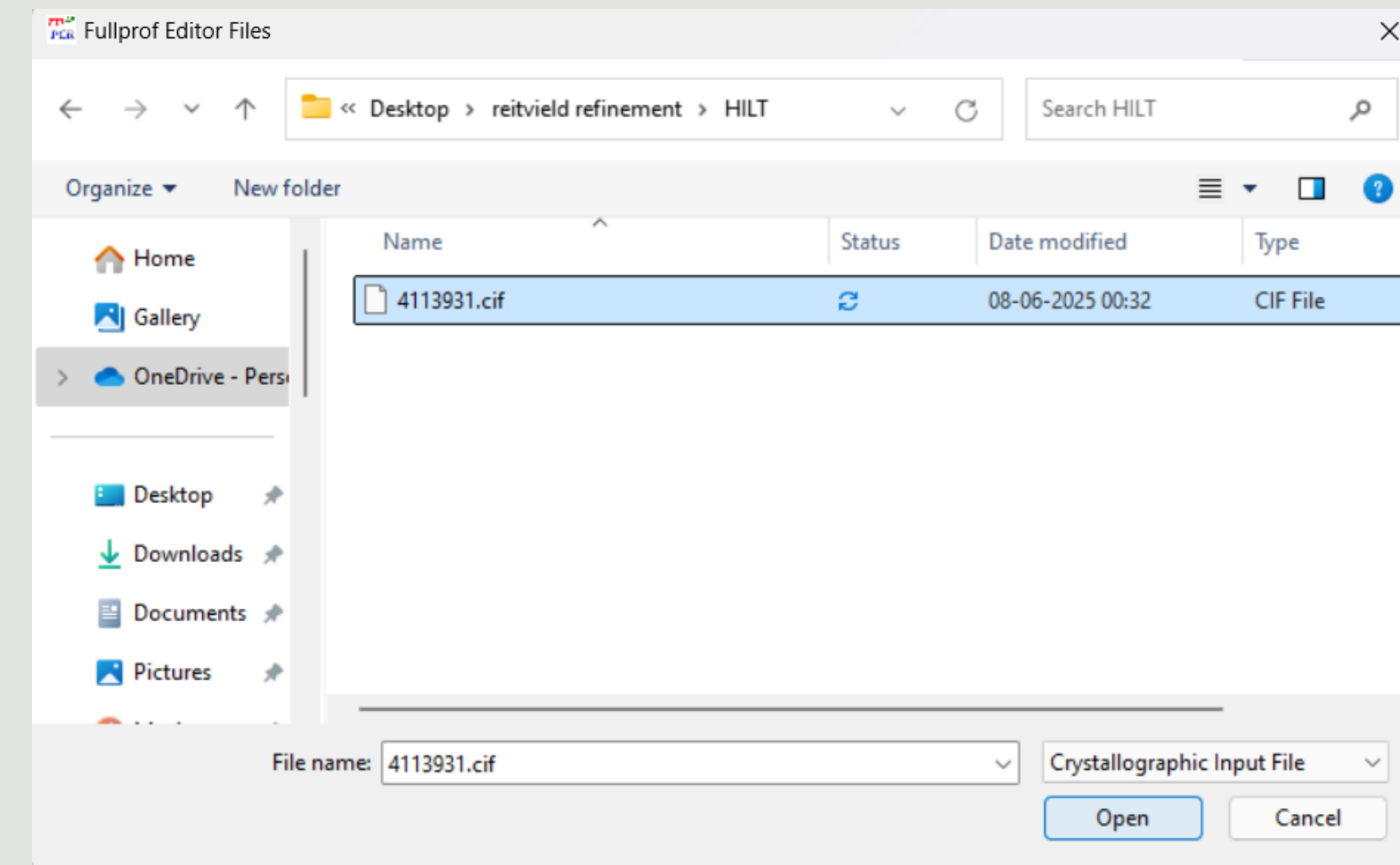
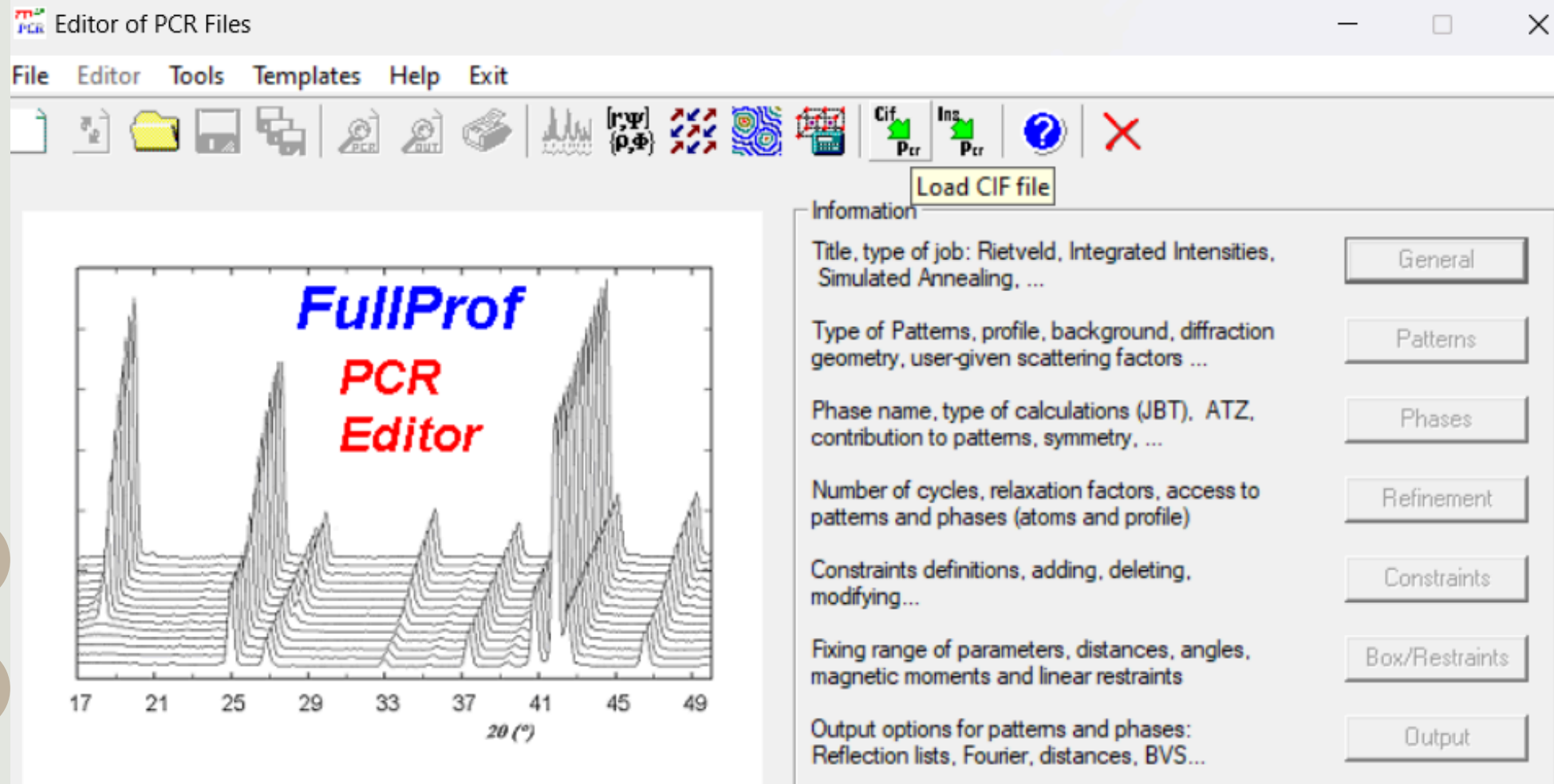
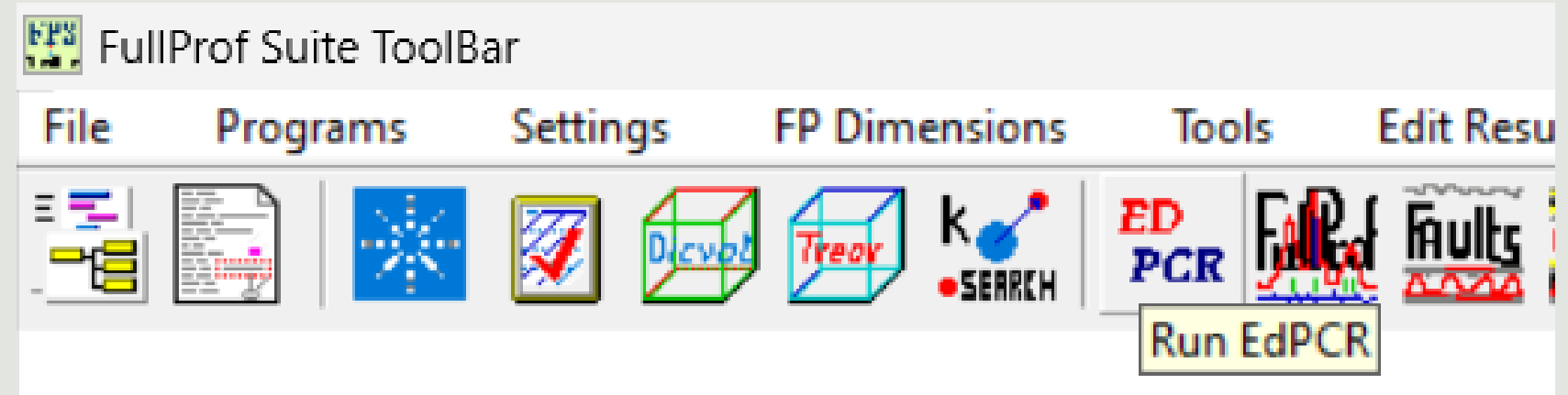
STEP-WISE RIETVELD REFINEMENT PROCESS

- Prepare Input Files

- Gather:
 - Crystallographic data: .cif (from COD)
 - Experimental data: data.dat
 - Background: background.bgr

- Create PCR File

- Launch **EdPCR** (FullProf GUI).
- Use CIF to PCR tool: Import .cif → Auto-generate initial .pcr file.



STEP-WISE RIETVELD REFINEMENT PROCESS

- Key Sections in EdPCR:
 - **General:** select refinement/calculation of a powder diffraction profile.
 - **Pattern:** in this select *data file/peak shape*:
 - select format as free format(2ThetaI,step,2ThetaF).

Profile Data Information: Pattern 1

Data File / Format | Refinement / Simulation | Pattern Calculation/Peak Shape

Data File: C:\Users\Acer\OneDrive\Desktop\rietveld refinement\HILT\4113931 Browse...

Format

- ☐ D1A/D2B (Old Format)
- ☒ Free Format (2thetaI, step, 2ThetaF)
- ☐ Variable Time X-ray Data
- ☐ D1A/D2B/3T2/G42
- ☐ Two Axis Instrument, G41
- ☐ X,Y,SIGMA (XYDATA)
- ☐ D1B (Old Format)
- ☐ GSAS Format
- ☐ X'Celerator (PANalytical)
- ☐ D1B/D20
- ☐ Socabim Software
- ☐ ISIS multi-bank normalized
- ☐ D4/D20L
- ☐ Synchrotron (Brookhaven)
- ☐ Synchrotron (DBWS Software)
- ☐ DMC/HRPD (P.S.I.)

OK Cancel

General Information

Title: Structural Tuning of Charge, Orbital, and Spin Ordering in Double-Cell

Calculations

- ☒ Refinement/Calculation of a Powder Diffraction Profile
- ☐ Refinement on Single Crystal Data / Integrated Intensity Data
- ☐ Simulated Annealing Optimization (Integrated Intensities) S.A. Options

☐ Optimize calculations according to the particular options used in this Job

OK Cancel

Patterns Information

Information

Pattern: 1/1 Weight: 1.0000

Initial Previous Add Del Next Last

Data file/Peak shape
Background Type
Excluded Regions
Geometry/IRF
User Scatt. Factors

OK Cancel

STEP-WISE RIETVELD REFINEMENT PROCESS

- Key Sections in EdPCR:

- **Pattern:**

- select simulation as XRay.
- select peak shape as pseudo-voigt then scattering variable 2Theta and then manually set the range of 2Theta.

Profile Data Information: Pattern 1

Data File / Format | Refinement / Simulation | Pattern Calculation/Peak Shape

Simulation / Refinement Data

☒ X-Ray ☐ Pattern Calculation (X-Ray)

☐ Neutron - CW (Nuclear and Magnetic) ☐ Pattern Calculation (Neutron - CW)

☐ Neutron - T.O.F (Nuclear and Magnetic) ☐ Pattern Calculation (Neutron - T.O.F.)

Wavelength

Cu λ_1 1.540560 λ_2 1.544390 (I_2 / I_1) 0.5000

OK Cancel

Profile Data Information: Pattern 1

Data File / Format | Refinement / Simulation | Pattern Calculation/Peak Shape

Peak Shape

Pseudo-Voigt ☒ Codefil.SHP ☐ Global.SHP

Scattering Variable

☒ 2Theta ☐ T.O.F. (microseconds) ☐ Energy (keV)

Range

Theta_min: 20.0000 Theta_max: 100.0000 Step: 0.0100

Range of calculation of a single reflection in units of FWHM: 8.0000

Incident beam angle at sample surface (°): 0.000

OK Cancel

STEP-WISE RIETVELD REFINEMENT PROCESS

- Key Sections in EdPCR:

- **Pattern:** now select background type :

- select linear interpolation between a set background points with refinable heights.

- **Phases:** name the phase , select the provided by user option .

- select contribution to patterns.

Background Information

Background Mode

☐ 6-Coefficients polynomial function

☐ 12-Coefficients polynomial function

☐ Debye-like (12-coeff.)+ polynomial functions (6-coefficients)

☐ 12-Coefficients Fourier-cosine series

☐ Fourier Filtering

☐ Background File transformed by 4-coefficients expression

Origin of the polynomial:

Number of points taken for Fourier Filter:

Browse...

☒ Linear Interpolation between a set background points with refinable heights

☐ Interpolation by cubic splines

☐ Chebychev Polynomial (24 coefficients)

OK Cancel

Phase Information: Phase 1

General Information on Phases

Name of Phase :

Calculation:

Coefficient to calculate the weight percentage of the Phase: ☐ Calculated automatically ☒ Provided by user

Contribution to patterns, preferred orientation direction, reflection list, ...

Space Group symbol/number, symmetry operators, basis functions, etc

OK Cancel

STEP-WISE RIETVELD REFINEMENT PROCESS

○ Key Sections in EdPCR:

■ Phases:

- select the type of pattern as XRay, peak shape as pseudo-voigt .
- select symmetry and set symm op automatic.

Symmetry Information

Space Group Properties

Symmetry Operators:

Spacegroup:

Symmetry operators | Magnetic/Displacement Operators | Irreducible representations

Laue Class: ☒ Centrosymmetric Case

Number of Symmetry Operators:

Num	Symmetry	TR	Num	Symmetry	TR
1	x,y,z	☐	2	x,-y,-z	☐
3	-x,y,-z	☐	4	-x,-y,z	☐
5	y,z,x	☐	6	-y,-z,x	☐

TR=Time reversal associated to symmetry operator

☐ Time Reversal for Inversion operator

Pattern Contribution Information for Phase 1

Pattern 1 | Pattern 2 | Pattern 3 | Pattern 4 | Pattern 5 | Pattern 6 | Pattern 7

☒ Current Phase contributes to the pattern

Type of Pattern

☒ X-Ray ☐ Pattern Calculation (X-Ray)

☐ Neutron (Constant Wavelength) ☐ Pattern Calculation (Neutron - Constant Wavelength)

☐ Neutron (T.O.F) ☐ Pattern Calculation (Neutron - T.O.F.)

☐ Nuclear and Magnetic

Peak Shape

☒ Codefil.shp ☐ Global.shp

Intensities

Reflection list:

☐ Use special control of parameters for peak overlap, rejected reflections for current phase

Brindley coefficient:

Global weight of the integrated intensity data vs profile data:

Factor for excluding reflections [$I < \text{Factor} * \text{Sigma}(I)$]:

Weights are divided by reduced χ^2 of precedent cycle:

STEP-WISE RIETVELD REFINEMENT PROCESS

- Key Sections in EdPCR:
 - **Refinement:** set cycles of refinement 20-50 as required , select reflection order as each cycle then select background.
 - import the .bgr file .

Refinement Information

Cycles of Refinement: 50

Stop Criterium of Coverage

Forced Termination when shifts < 0.10 x E.S.D.

Others: None

Relaxation Factors for Shifts

Atomic 1.00 Anisotropic 1.00 Profile 1.00 Global 1.00

Reflections ordering

☐ Only at the first cycle ☒ Each cycle

☐ Bragg R-Factor excluding reflections limiting excluded regions

Pattern 1 | Pattern 2 | Pattern 3 | Pattern 4 | Pattern 5 | Pattern 6

Refinement weighting model

☒ Least Squares

☐ Maximum Likelihood

☐ Unit Weights

Background

Instrumental

Micro-Absorption

Reduction factor of number of data points: 0

OK

Cancel

Phase 1 | Phase 2 | Phase 3 | Phase 4 | Phase 5 | Phase 6

Atoms

Prop. Vectors

Patterns

☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Profile

Micro-Structure

HKL Shifts

Further Parameters

Linear interpolation between a set of Background Points: Pattern 1

Interpolation Method

☒ Linear Interpolation ☐ Cubic Splines Interpolation

Information

Number of Points: 32

	2Theta	Counts	
1	24.356	188.194	☐
2	26.337	182.793	☐
3	28.883	209.794	☐
4	31.523	234.094	☐

Refine All

Fix All

Import from Background File

OK

Cancel

STEP-WISE RIETVELD REFINEMENT PROCESS

- Initial Refinement Setup

- In PCR file, deactivate all parameters initially .
- Activate only:
 - Scale factor
 - Background coefficients (if using polynomial)
 - Zero-point shift (2 θ correction)

- Iterative Refinement

- Cycle 1:
 - Refine scale + background + zero shift.

Profile Parameters: Phase 1 Pattern 1

Factors

	Scale		Overall B-factor	
Coefficients	0.10000E-02	☐	0.0000	☐

Cell Parameters

	a		b		c		alpha		beta		gamma	
Coefficients	2.862840	☐	2.862840	☐	2.862840	☐	90.000	☐	90.000	☐	90.000	☐

FWHM / Shape Parameters Asymmetry Parameters Preferred Orientation

FWHM Parameters

	U		V		W		IG	
Coefficients	0.004133	☐	-0.007618	☐	0.006255	☐	0.000000	☐

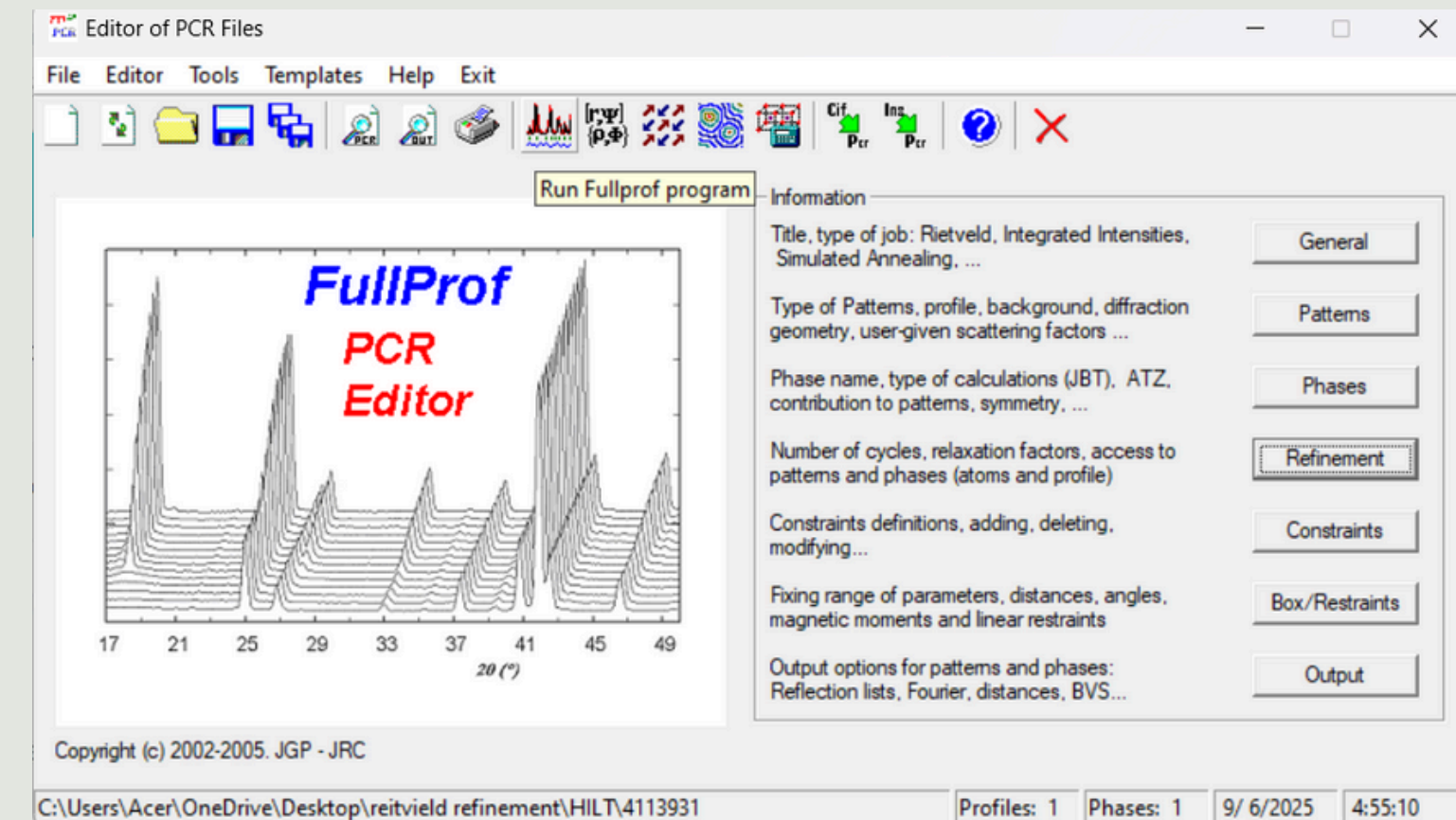
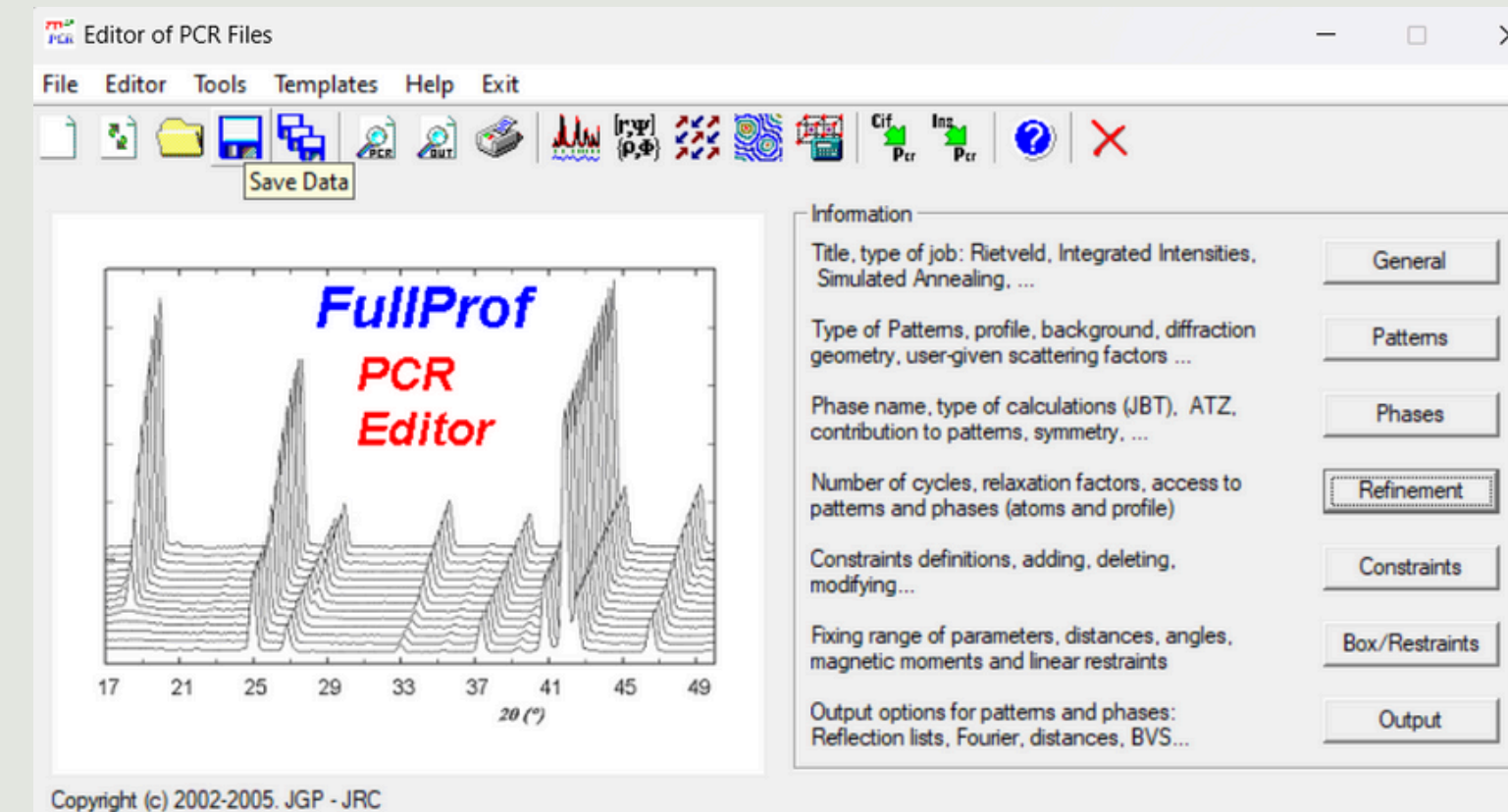
Shape Parameters

	Eta_0		X				
Coefficients	0.000010	☐	0.018961	☐	☐	☐	☐

☐ Refine FWHM for second wavelength

	U2		V2		W2	
Coefficients		☐		☐		☐

Refine All
Fix All
Cancel
OK



STEP-WISE RIETVELD REFINEMENT PROCESS

- **Iterative Refinement**

- Cycle 2:
 - Activate unit cell parameters. Re-run.
- Cycle 3:
 - Activate peak shape parameters (U, V, W, η). Re-run.
- Cycle 4:
 - Activate atomic coordinates (x, y, z) and occupancies. Re-run.
- Cycle 5:
 - Activate isotropic/anisotropic displacement parameters (Biso/Uij).

- **Monitor Refinement**

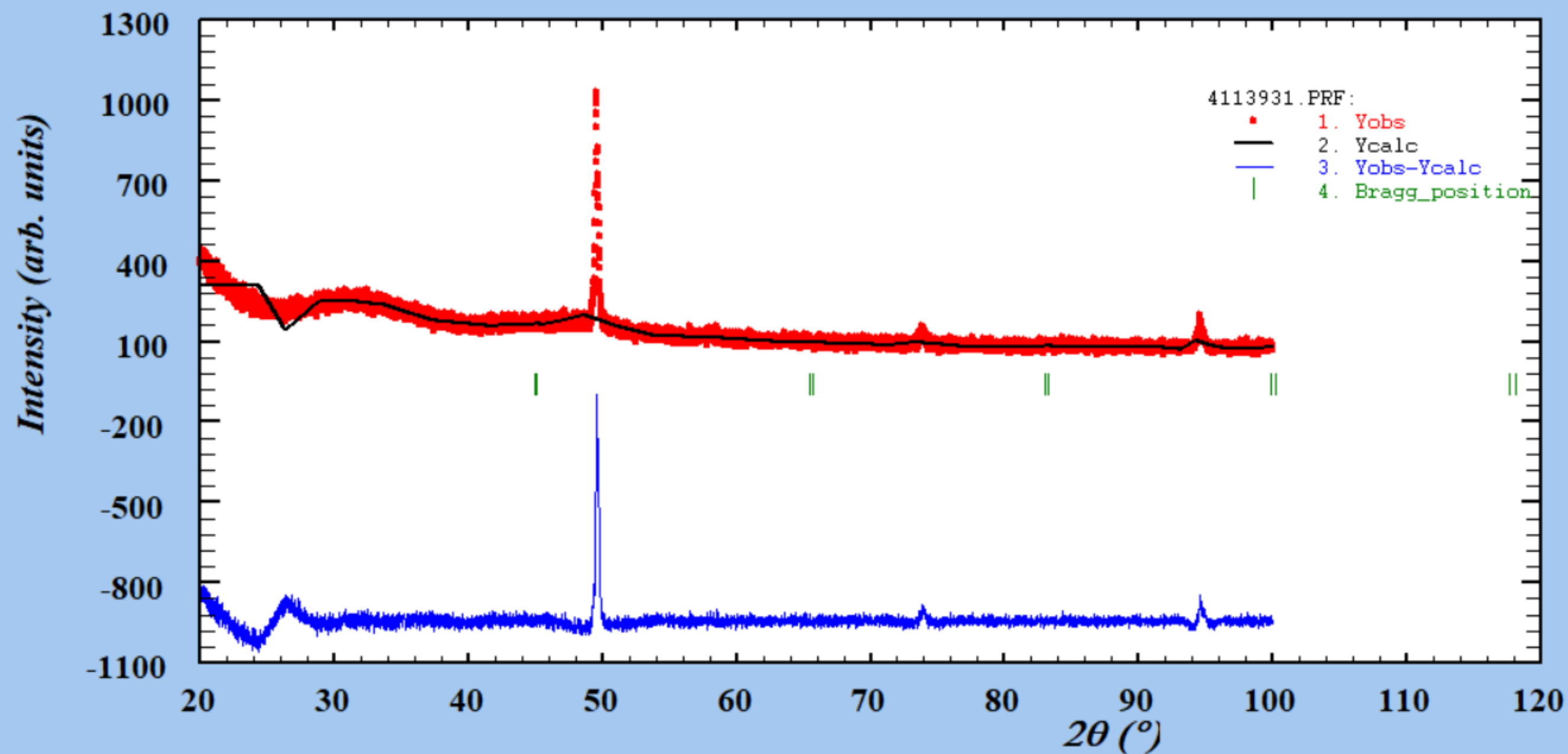
- After each cycle:
 - Check χ^2 reduction (aim for $\chi^2 < 2-3$).
 - Examine residual plot (difference curve) for misfits.
 - Ensure parameter shifts are $< 10\%$ of their errors.

- **Final Steps**

- Save final .pcr and .out files.

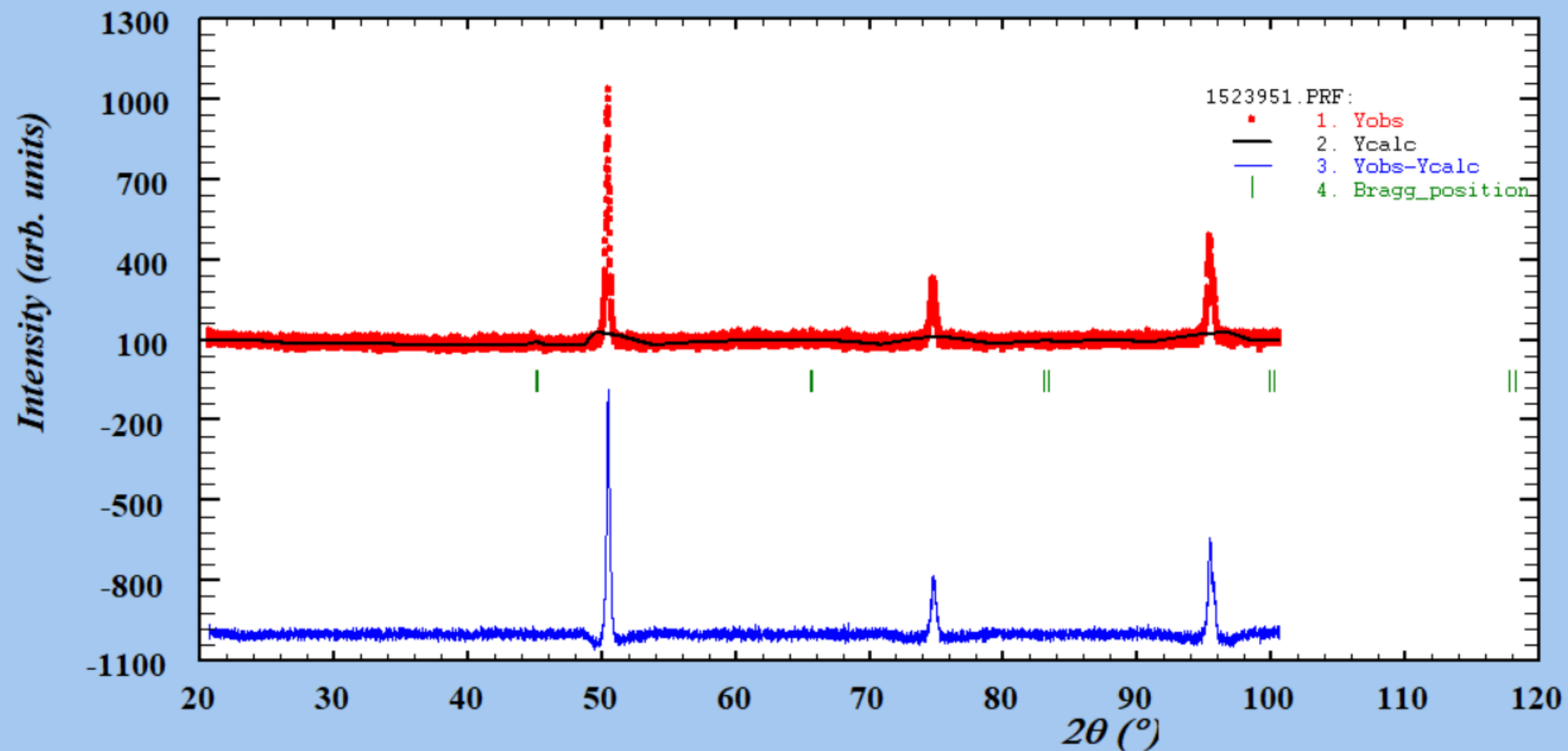
RESULT (HILT)

Structural Tuning of Charge, Orbital, and Spin Ordering in Double-Cell Chi2: 4.1542



RESULT (SURFACE)

The lattice parameters and solubility limits of alpha-iron as affected by Chi2: 5.8094





Thank You

For your attention